

Firefox OS: Web to the Core

>>There's yet another smartphone OS joining the already confused selection in 2013, and that is Firefox OS. Mozilla's latest endeavour is an operating system built entirely for the web, sound familiar?> by Kshitij Sobti

Firefox OS sounds a lot like Google's Chrome OS, down to the fact that both operating systems are named after their respective browsers. Like Chrome OS which is a Chrome-based browser, Firefox OS is a Firefox-based OS. While similar, they are both actually quite different, and have different goals. What Firefox OS is actually a lot closer to is HP / Palm's webOS.

Firefox OS is an operating system that is designed to run only web applications. On the surface this sounds a lot like Chrome OS and the difference between the OSs seems to be that Chrome OS is for notebooks while Firefox OS is for mobile. However the deeper difference becomes apparent when you consider that in Chrome OS the browser, the UI, and everything are all native code running on Linux; Firefox OS on the other hand is an operating system that itself is a web app!

Firefox OS is built entirely using web technologies, this means even the home screen, dialer and even the browser in Firefox OS is written in HTML, CSS, and JavaScript.

Firefox OS's architecture consists of three layers which Mozilla has named Gonk, Gecko and Gaia. The middle one, Gecko might sound familiar, stick with us and you will understand why. Here is what these layers are and what they do:

Gaia: This is the visible layer of the OS, it is the UI that you see when you run Firefox OS. Gaia includes all the basic apps of Firefox OS including the dialler, SMS app, the camera app, home screen, lock screen, OS setting app, and almost everything else you would consider an essential part of a mobile OS. Since this is simply a web application, you can easily run it on Firefox, or even Chrome on a desktop or a modern browser running on any platform.

Gecko: Gecko is actually the rendering engine used by Firefox. This is what Gaia runs on. Think of it as a fullscreen browser running on the mobile device, with everything else running on top of this full screen browser. The difference between this version of Gecko, and the Gecko have running in desktop or mobile versions of Firefox is that the Firefox OS version of Gecko has mobile APIs required for mobile web applications enabled.

Gonk: Gonk is the lowermost layer of Firefox OS. This consists of the Linux kernel and other userspace apps necessary to run the Firefox engine. Since this layer is developed by Mozilla it can give Gecko more access to native hardware than is available on any other OS. This layer actually shares a lot with Android since it has a similar goal of running on mobile devices.



Firefox OS is a web app running on the Firefox engine on Linux. An entire mobile OS written in web technologies that can install and manage other web applications in itself sounds like a big accomplishment, but it is only a small part of the whole. After all, how does a phone even work, how can a web app make calls?

The web platform is the perfect platform for creating cross-platform applications. After all an app can be written once, using one well known technology and it will run on every device. So why do people still spend so much time and money making native apps for different platforms. The fact is that while web apps can reach far, they cannot reach deep. They cannot use features such as system's camera, or Bluetooth. They cannot use NFC or telephony features of a mobile system.

What is left for a mobile app to do?

When you think of the SMS app in a mobile, why does it seem like it's something that a web app could not do? Microsoft can bring its office suit with complex rich formatting and a ribbon interface to the web, but a simple text message sending application seems out of reach. The only thing missing is the last piece of the puzzle, access to the system functionality of sending a message.

Currently many developers use toolkits such as PhoneGap to bridge the functionality gap between the web and native apps. However due to the work done by Mozilla and others this might not